DISTANCE EDUCATION

B.Sc. (Computer Science) DEGREE EXAMINATION, MAY 2012.

DIGITAL COMPUTER FUNDAMENTALS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Convert the following :

   (i) $(1000)_{10} = (?)_8$

   (ii) $(0.5632)_{10} = (?)_8$. (10)

   (b) Write the procedure for converting decimal float number to its binary equivalent. (10)

2. (a) Convert the following :

   (i) $(5555)_2 = (?)_{16}$

   (ii) $(0.0111)_2 = (?)_{16}$. (10)

   (b) How to convert integer number to its octal equivalent? (10)

3. (a) Explain the construction of ALU. (10)

   (b) Explain how to represent the positive and negative integer number for addition operations. (10)

4. (a) How Quine Mckhuskey method can be used to reduce the Boolean expressions? (10)

(b) With an example, show how to simplify the sum of products expressions and products of sums expression. (10)

5. (a) Mention the fundamental axions of Boolean Algebra. (10)

(b) Simplify the expressions :

$ABCD + ABC\overline{D} + \overline{A}BCD + \overline{A}BC\overline{D}$. (10)

6. (a) Explain the working concept of parallel Binary Adder. (10)

(b) Explain the function of Half adder with an diagram. (10)

7. (a) Explain the RS flipflop with its diagramatic representation. (10)

(b) Give the purpose of binary counters and BCD counters. (10)

8. (a) Describe the Data representation with all data types. (10)

(b) Discuss the classification of computer generation. (10)

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DISTANCE EDUCATION

B.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2012.

DATA STRUCTURES USING C

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) What are the different types of constants available in C? Explain with example. (10)

   (b) How to declare variable in the program. Explain with example program. (10)

2. (a) Explain different types of operators with example. (10)

   (b) Explain different types of if statements. (10)

3. (a) What is the use of function? Explain. (10)

   (b) How to pass arguments to a function? Explain with example. (10)

4. Write notes on :

   (a) Opening and closing a data file. (7)

   (b) Creating a data file. (5)

   (c) Processing a data file. (8)

5. (a) How to implement stack structure in you program. Explain with example. (10)

(b) Explain "Queue" operations in detail. (10)

6. (a) Explain with example linked list. (10)

(b) Explain doubly linked list. (10)

7. Write notes on :

(a) Stack operations. (10)

(b) Queue operations. (10)

8. (a) How to represent a binary tree. Explain with your own program. (10)

(b) Briefly explain singly linked list with example. (10)

DISTANCE EDUCATION

B.Sc. (Computer Science) DEGREE EXAMINATION, MAY 2012.

DISCRETE MATHEMATICS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Show that $P \to (Q \to P) \Leftrightarrow \neg P \to (P \to Q)$. (7)

   (b) Show that $Q \vee (P \wedge \neg Q) \vee (\neg P \wedge \neg Q)$ is a tautology. (7)

   (c) Construct the truth table for $(P \to Q) \rightleftarrows (\neg P \vee Q)$. (6)

2. (a) Show that $(A - B) - C = (A - C) - (B - C)$. (6)

   (b) Define reflexive, symmetric, transitive, irreflexive and antisymmetric. (8)

   (c) Let $S = \{\langle x, x^2 \rangle / x \in N\}$ and $T = \{\langle x, 2x \rangle / x \in N\}$ where $N = \{0, 1, 2, ...\}$, find the ranges of S and T. (6)

3. (a) Let $f : x \to y$ and $g : y \to z$ be any two functions. If f and g are both one-to-one onto, prove that $g \circ f$ is also one-to-one, onto. (10)

   (b) Show that the function $f\langle x, y \rangle = x + y$ is primitive recursive function. Hence compute the value of $f\langle 2, 3 \rangle$. (10)

4. (a) Define semigroup and monoid with examples. (6)

(b) Let $\langle S,*\rangle, \langle T,\Delta\rangle$ and $\langle V,\oplus\rangle$ be semigroups and $f:S\to T$ and $g:T\to V$ be semigroup homomorphism. (6)

(c) In any group $\langle G,*\rangle$, prove that $(a*b)^{-1}=b^{-1}*a^{-1}$ for all $a,b\in G$. (8)

5. (a) Prove that the order of a subgroup of a finite group divides the order of the group. (7)

(b) Let $f:G\to G'$ be a group homomorphism from the group $(G,*)$ to $(G',0)$. Prove that

(i) $f(e)=e'$

(ii) $f(a^{-1})=f(a)^{-1}$. (6)

(c) If $f:G\to G'$ is a group homomorphism, prove that its Kernel K is a normal subgroup in G. (7)

6. (a) Define digraph, complete graph, bipartite graph, elementary path and geodesic. (10)

(b) Prove that in any graph, the number of vertices of odd degree is even. (5)

(c) Show that in a simple digraph $G=\langle V,E\rangle$, every node of the digraph lies in exactly one strong component. (5)

7. (a) Let G be an undirected graph. Prove that G is bipartite if and only if it contains no add cycle. (10)

(b) Prove that a simple graph with n vertices and k components can have at most $(n-k)(n-k+1)/2$ edges. (10)

8. (a) Show that $A \cap (B - C) = (A \cap B) - (A \cap C)$. (8)

(b) Let $f : A \to B$ and $g : B \to C$ be both one-to-one and onto function. Prove that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. (6)

(c) Prove that the relation "congruence modulo m" over the set of positive integers is an equivalence relation. (6)

DISTANCE EDUCATION

B.Sc. (Computer Science) DEGREE EXAMINATION,

MAY 2012.

COMPUTER LAB I — C AND DATA STRUCTURES

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the students by lot system.

Answer BOTH subdivisions (a) and (b).

1. (a) Write a C program to find square root, square and cube of any number.

   (b) Write a C program to evaluate the given mathematical expression using stack.

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2. (a) Write a C program to input a temperature in Celsius and find the corresponding temperature in Fahrenheit. Use the formula F = 9/5 * C + 32.

   (b) Write a C program to implement insert and delete operations on queue using array concept.

3. (a) Write a C program to compute commission earned by a salesman according to the scheme given below.

Sales Amount	Commission %
Upto Rs. 1,000	0
Above 1000 upto 5000	5
Above 5000	10

(b) Write a C program to sort 10 No's in ascending order with naming of variable and the value before and after sorting.

---------- Cut here ----------

4. (a) A man is paid at the hourly rate of R. 15 per hour for the first 45 hours worked. Thereafter, overtime is paid at 1.5 times the hourly rate for the next 25 hours and 2 time the hourly rate for further hour worked per week, calculate and print his gross weekly wage.

(b) Write a C program to sort a set of elements using Insertion sort.

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5. (a) Write a C program to concatenate two given strings and find the length of the concatenated string.

(b) Write a C program for linked list implementation of queue operations.

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DISTANCE EDUCATION

B.Sc. (C.S)/B.Sc. (C.S)/(Lateral) DEGREE EXAMINATION, MAY 2012.

OBJECT ORIENTED PROGRAMMING AND C++

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions. ($5 \times 20 = 100$)

All questions carry equal marks.

1. (a) Discuss the basic concepts of' Object Oriented Programming. (10)

   (b) Illustrate the manipulators with example. (10)

2. (a) Explain the data types available in C++. (10)

   (b) Write a brief note on objects and classes. (10)

3. (a) What is a constructor? Explain with example. (10)

   (b) Write a C++ program to explain inline function. (10)

4. (a) How do you use static data and member functions? Explain. (10)

   (b) Discuss the fundamental aspects of pointers. (10)

5. (a) What is the use of 'this' pointer? Give an example. (10)

   (b) Describe the string functions in C++. (10)

6. (a) Write short notes on dynamic and copy constructors. (10)

   (b) What is meant by polymorphism? Explain. (10)

7. (a) Write a C++ program to demonstrate function overloading. (10)

(b) Explain the basic concepts of operator overloading. (10)

8. (a) Discuss the types of inheritance. (10)

(b) Explain about virtual functions with example. (10)

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (C.S) (Lateral) DEGREE EXAMINATION, MAY 2012.

SCIENTIFIC COMPUTING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Solve the system of equations by Gauss-Jordan method

$x + 2y + z = 3$; $2x + 3y + 3z = 10$; $3x - y + 2z = 13$.

(b) Solve the following system of equations by using Gauss-Seidal method (correct to 3 decimal places)

$8x - 3y + 2z = 20$; $4x + 11y - z = 33$; $6x + 3y + 12z = 35$.

2. (a) Find the values of y at $x = 46$ and $x = 63$ from the following data :

x :	45	50	55	60	65
y :	114.84	96.16	83.32	74.48	68.48

(b) Using Lagrange's interpolation formula, find $y(2)$ from the following table

x :	0	1	3	4
y :	5	6	50	105

3. (a) By dividing the range into ten equal parts, evaluate $\int_0^{\pi} \sin x \, dx$ by Trapezoidal rule and Simpson's rule.

(b) Evaluate $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x = 2$, given

x :	0	1	3	6
y :	18	10	–18	40

4. (a) Find the median and mean deviation of the following data :

Size :	0–10	10–20	20–30	30–40
Frequency :	7	12	18	25
Size :	40–50	50–60	60–70	
Frequency :	16	14	8	

(b) Calculate the median for the following frequency distribution :

Marks :	5–10	10–15	15–20	20–25	25–30
No. of students :	7	15	24	31	42
Marks :	30–35	35–40	40–45	45–50	
No. of students :	30	26	15	10	

5. (a) Fit a straight line trend for the following series. Estimate the value for 2004

Year :	1997	1998	1999	2000	2001	2002	2003
Production of steel : (in. tonnes)	60	72	75	65	80	85	95

(b) A box contains 3 red and 7 white balls. One ball is drawn at random and in its place a ball of the other colour is put in the box. Now one ball is drawn at random from the box. Find the probability that it is red.

6. (a) The probability density function $f(x) = Cx^2(1-x)$; $0 < x < 1$ find

(i) the constant C

(ii) $P\left(X < \frac{1}{2}\right)$.

(b) Find the standard deviation of the following distribution :

Age :	20–25	25–30	30–35	35–40	40–45	45–50
No. of persons :	170	110	80	45	40	35

7. (a) The life time of electric bulbs for a random sample of 10 from a large consignment gave the following data :

Item :	1	2	3	4	5	6	7	8	9	10
Life in '000 hours :	4.2	4.6	3.9	4.1	5.2	3.8	3.9	4.3	4.4	5.6

Can we accept the hypothesis that the average life time of bulbs is 4,000 hours?

(b) In an anti a malarial campaign in a certain area, quinine was administered to 812 persons out of a total population of 3,248. The number of fever cases is shown below.

Treatment	Fever	No Fever	Total
Quinine	20	792	812
No Quinine	220	2,216	2,436
Total	240	3,008	3,248

Discuss the usefulness of quinine in checking Malaria.

8. (a) Two random samples were drawn from two normal populations and their values are :

A :	66	67	75	76	82	84	88	90	92		
B :	64	66	74	78	82	85	87	92	93	95	97

Test whether the two populations have the same variance at the 5% level of significance. ($F = 3.36$) at 5% level for $v_1 = 10$ and $v_2 = 8$.

(b) Let X be a continuous random variable with p.d.f. given by $f(x) = \begin{cases} 2e^{-2x} \; ; & x > 0 \\ 0 \; ; & x \geq 0 \end{cases}$

Find the mean and variance.

DE–3635 **25**

DISTANCE EDUCATION

B.Sc. (C.S.)/ B.Sc. (C.S.) (Lateral) DEGREE EXAMINATION, MAY 2012.

APPLICATION PROGRAMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions. ($5 \times 20 = 100$)

All questions carry equal marks.

1. (a) Write down the steps to copy files from one folder to another. (10)

   (b) Explain any five buttons on the standard toolbar. (10)

2. (a) Write a short note on screen saver settings. (10)

   (b) Explain the different ways of page numbering in a document in word. (10)

3. (a) What are the different ways to create a document in word? Explain. (10)

   (b) Write a note on spellcheck and mailmerge features in word. (10)

4. (a) How to enter formula in the formula bar? Explain (10)

   (b) What are the different types of charts in Excel? Explain. (10)

5. (a) Explain Find and Replace in Excel. (10)

(b) How will you create and manipulate slides in Powerpoint? (10)

6. (a) Explain the process of insert, delete and copy slides using Powerpoint. (10)

(b) Describe the different slide layout in Powerpoint. (10)

7. (a) Explain about the different views in Powerpoint. (10)

(b) How do you create table in Ms-Access? Explain entering and editing data in a table. (10)

8. (a) Write a short notes on creating a report in Access. (10)

(b) Explain the database creation in Ms-Access. (10)

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, MAY 2012.

COMPUTER LAB II (C++ PROGRAMMING LAB)

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the student by lot system.

Each subdivision carries 50 marks.

1. (a) Using dynamic constructors write a C++ program to concatenate two given strings.

   (b) Create a C++ class for stock item abstract data type. It should have the attributes of stock levels (an integer) and unit price (a float). Define the methods to return the values of these two attributes and to set them using parameters. Add two more methods to allow stock receipts and issues updating the stock level as appropriate. Write a menu driven program to implement the above problem.

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2. (a) Using pointers create a class and write a program to get the n names and display them in sorted order.

   (b) Write a program using friend function frifuc ( ) which can act on the classes alpha and beta. Using constructors fix the values for alpha and beta.

3. (a) Using operator overloading , write a C ++ program to find the difference and total length of given two various tubes specified is meters and centimeters.

(b) Write a program that emulates the DOS copy command. That is, it should copy the contents of characters file (such as any CPP file) to another file. Invoke the program with two command line arguments-the source file and the destination file.

C > Copy srsfile. ext destfile. ext.

In the program , check that the user had typed the correct number of command line arguments and that the files specified can be opened. Improve on the DOS TYPE command by having the program signal an error if the destination file already exists.

---------- Cut here ----------

4. (a) Write a menu driven C++ program to add and subtract given two matrices of order $m \times n$ defined in class, using operator overloading.

(b) Write a program using polymorphism to calculate the square of any two numbers of type int, float, double and long.

---------- Cut here ----------

5. (a) Create a class employee that contains employee number, name and address. Write a menu driven C++ program to get the 'n' number of employee details and display all details in employee name wise sorted order.

(b) Create a equivalent of a four-function calculator. The program should request the user to enter a number, an operator , and another number. It should then carry out the specified arithmetic operations: adding, multiplying , subtracting and dividing the two numbers. Finally it should display the result. When it finishes the calculation, the program should ask if the user wants to do another calculation.

6. (a) Create a class called employee that contains a name and an employee number. Include a member function called getdata ( ) to get data from the user, another function called putdata ( ) to display the data. Write a main ( ) program to exercise this class. It should create an array of type employee and then invite the user to input data for 'n' employees.

(b) Write a C++ program which will accept a string of n characters in length from the keyboard and count the occurrences of each of the five vowels in the string. The output should be in a (tapped) format similar to this example

A	E	I	O	U
5	2	0	4	2

---------- Cut here ----------

7. (a) Write a program to process students mark with the help of classes. The class has private variables for name, mark 1, mark 2 and mark 3. It has two member functions – getdata ( )- to get input-result ( )- to print the results. All subject mark must be > = 50 for pass otherwise fail.

(b) Write a function called reverseit ( ) that reverse a string. Use for loop that swaps the first and last characters, then the second and next - to -last characters and so on. The string should be passed to reverseit ( ) as an argument. Write a program to exercise reverseit ( ). Test the program with Napolean's famous phrase " Able was I ere I saw Elba".

8. (a) Create a class DONOR that contains donor name, donor number , age, address , sex, blood group. Write a menu driven C++ program to display the number , name and address of the donors for the following categories.

(i) blood donors having the blood group A+

(ii) blood donors in the age group between 30 to 50.

(iii) female donors having blood groups O+ in the age between 16 to 25.

(b) Write a program that reads a group of numbers from the user and places them in an array of type float. Once the numbers are sorted, the program should average them and print the result. Use pointer notation whenever possible.

DISTANCE EDUCATION

B.Sc.(Computer Science)/B.Sc.(CS) (Lateral) DEGREE EXAMINATION, MAY 2012.

COMPUTER — LAB III (APPLICATION PROGRAM LAB)

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and is given to the candidate by LOT system.

Each subdivision carries 50 marks.

1. (a) Prepare your resume. Use table, bullets and different colour features.

   (b) Create two tables SALES 1 and SALES 2. Joint them to produce a third table SALES 3.

---------- Cut here ----------

2. (a) Prepare the first page of MCA practical record note book with picture insertion and alignment.

   (b) Draw Line, bar, pie charts for the data given below:

ABC Company Ltd.

Year	Sales	Cost	Profit
1991	1000	400	600

Add data for 10 years from 1991 to 2000. Provide titles, legends, grids and data labels.

3. (a) Type lecture notes and provide audio explanation with the help of sound files.

(b) Create STUDENT table with fields register number, internal and external marks for five subjects and table SUBJECT with fields subject code and subject name for five subjects. Create a report in the following format:

ALAGAPPA UNIVERSITY, KARAIKUDI

STATEMENT OF MARKS

REG NO. ..... NAME : .....

SUB CODE	SUBJECT NAME	INTERNAL	EXTERNAL	TOTAL	RESULT
--	--	--	--	--	--

(i) Condition for passing a subject: A minimum of 35 marks in external and 50 marks in total.

(ii) Display FAIL in red colour.

---------- Cut here ----------

4. (a) Prepare an invitation for a function to be conducted in your institution. Use different text orientation and pictures to make it attractive.

(b) The following were the observations made in certain experiments for the values y and given the values of x

x :	1	2	3	4	5	6	7	8	9	10
y :	10	30	45	25	15	28	40	32	15	35

Compute the following:

(i) Find the mean, median and standard deviation

(ii) Draw the bar chart and shading cells.

(iii) Draw a chart in Excel and paste it in word.

5. (a) Prepare a news report using two columns, insert a picture in the first column and make the text flow around it.

(b) Create employee table with the fields:

employee number, name, age, sex, street, city, pin and salary.

(i) Add data for 10 employees

(ii) Write a query to display all male employees whose salary is between 1000 and 5000 and living in city "COIMBATORE".

(iii) Show all the records in the table for the city "CHENNAI" by filtering

---------- Cut here ----------

6. (a) Create a form letter than informs the customer about the date of maturity of a deposit amount in a commercial bank and request the customer for renewal. Mail merge it with an access database.

(b) Create two worksheets containing day to day house hold expenses for the months July and August 2010. With the following columns:

ITEM	AMOUNT

(i) Add 10 different items for two months in two different worksheets.

(ii) Consolidate both the months data and find the total expense on each item.

(iii) Find the max and min expense amount.

DE–3636 **31**

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) (lateral) DEGREE EXAMINATION, MAY 2012.

COMPUTER GRAPHICS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

1. (a) Explain the simple DDA algorithm for drawing a line with an example. (10)

   (b) Explain the method of generating a circle with necessary equations. (10)

2. (a) What is region filling? Explain any one techniques used for region filling? (10)

   (b) Explain the three basic two dimensional transformations with an example. (10)

3. (a) Explain the two dimensional line clipping algorithm. (12)

   (b) Discuss the terms viewing and aspect ratio. (8)

4. (a) Explain three dimensional composite transformation with an example. (12)

   (b) Discuss three dimensional mirror reflection with an example. (8)

5. (a) Explain any one three dimensional clipping algorithm. (12)

(b) Discuss the concept of three dimensional projection and viewing. (8)

6. (a) Explain Z-buffer algorithm and its use. (10)

(b) Discuss the components of user interface. (10)

7. (a) What is command language? Explain. (10)

(b) Discuss the applications of computer graphics. (10)

8. Write notes on :

(a) Graphics system software

(b) Point clipping

(c) Video display generation. (7 + 7 + 6)

DISTANCE EDUCATION

B.Sc (CS)/B.Sc (CS) (Lateral) DEGREE EXAMINATION, MAY 2012.

VISUAL PROGRAMMING

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Define Windows Programming. How does it differ from traditional programming? (10)

   (b) Explain how to format a window. (10)

2. (a) Write notes on multi tasking. (10)

   (b) Explain the various operating modes on windows programming. (10)

3. (a) Explain briefly the different control structures in VB. (10)

   (b) Write short notes on menu and dialog boxes. (10)

4. (a) Write short notes on:

   (i) Device context (5)

   (ii) Windows messages. (5)

   (b) Explain the usage of paint brush. (10)

5. (a) Explain the various tools available in VB. (10)

(b) Define about Active X Controls. (10)

6. (a) Write notes on:

(i) OLE

(ii) MDI (10)

(b) Bring out the differences between sub procedures and function procedures. (10)

7. (a) What is meant by VB Module? Explain with an example. (10)

(b) How will you design menus in VB? (10)

8. (a) Define an exception. How it is handled? (10)

(b) What do you mean by procedure? State their benefits. (10)

DE–3638 **33**

DISTANCE EDUCATION

B.Sc. (C.S.)/B.Sc. (C.S.) (Lateral) DEGREE EXAMINATION, MAY 2012.

OPERATING SYSTEMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What is an operating system? Explain. (10)

   (b) Discuss in detail about system calls. (10)

2. (a) Explain about interprocess communication in detail. (10)

   (b) Discuss about round robin scheduling. (10)

3. (a) Write a detailed note on deadlock prevention and avoidance. (10)

   (b) Discuss the goals of I/O software. (10)

4. (a) Explain in detail about virtual memory. (10)

   (b) Discuss memory management without swapping or paging. (10)

5. (a) How to improve the reliability and performance in file management? (10)

   (b) Explain about file servers, security and its protection mechanisms. (10)

6. (a) Explain the concepts of operating system in brief.(10)

(b) Differentiate between process, processes and processors. (10)

7. (a) Write a detailed note on interrupt handlers. (10)

(b) Discuss on multiprogramming with fixed partitions. (10)

8. (a) Discuss on file system reliability and performance. (10)

(b) What is meant by policy driven scheduling? Explain with an example. (10)

DE–3639 **34**

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (Computer Science)(Lateral)

DEGREE EXAMINATION, MAY 2012.

INTERNET PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5× 20 = 100)

1. (a) Explain in detail about the TCP/IP.

   (b) Discuss about the Internet address protocols.

2. (a) Write notes on: (i) file transfer protocol. (ii) HTTP.

   (b) Describe in detail about HTTP methods.

3. (a) Discuss about the Name Server concepts.

   (b) Write a note on URL.

4. (a) What is an Internet? Explain any two topologies.

   (b) What are the Resource available in internet? Explain.

5. (a) Write the features of object oriented programming.

   (b) Explain the structure of Java program.

6. (a) What is operator? Explain any four types of operators in JAVA.

(b) Explain about the array with an example.

7. (a) What is constructor? Explain with an example.

(b) Discuss about the multiple inheritance with an example.

8. (a) Explain about the character extraction with example.

(b) Briefly explain about the searching and modifying string in a given text with example.

DE–3640 **35**

DISTANCE EDUCATION

B.Sc. (C.S.)/B.Sc. (C.S.) (Lateral) DEGREE EXAMINATION, MAY 2012.

RDBMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain the concept of schema and subschema. (10)

   (b) Discuss in detail about different user interfaces. (10)

2. (a) What is meant by data independence? How to achieve it? (10)

   (b) Explain the data definition facilities. (10)

3. (a) Discuss about the mapping operation. (10)

   (b) Explain the oracle architecture in detail. (10)

4. (a) How can you join multiple tables through queries. (10)

   (b) Discuss the elements of SQL language. (10)

5. (a) How a table is created and modified? (10)

   (b) Discuss about the languages supported by oracle precompiler. (10)

6. (a) Discuss the advantages of database management system. (10)

(b) Explain the concepts of relational model. (10)

7. (a) Explain about the types of databases. (10)

(b) Discuss about the basic query in oracle. (10)

8. (a) Discuss about embedded SQL statements. (10)

(b) What are the semantic issues that arise in relational model? Explain. (10)

DE–4359 **36**

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, MAY 2012.

COMPUTER LAB – IV
(VISUAL BASIC AND ORACLE LAB)

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and is given to the candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Write and test a VB program to display the day of the given date and covert upper case from lower case for the given world.

(b) Create a table mark with the following fields :

Reg. No, Name, Tamil, English, Maths, Biology.

(i) Add five records.

(ii) Show data fields Reg. No, Name and Maths alone on the screen.

(iii) Add the field total.

(iv) Compute the total.

(v) Show all the fields on the screen.

2. (a) Write a VB program to calculate simple interest or compound interest using three command buttons namely input, calculate, display.

(b) Create a table address with the following fields :

Name, street, city, pin.

(i) Add five records.

(ii) Display the structure of the table.

(iii) Add the field district.

(iv) Fill all district with chennai.

(v) Change the pin to 630020 where city is Trichy and pin is 636003.

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3. (a) Write a test a VB program to select candidates for four posts, based on their qualifications:

Male and PG degree	–	Manager
Male, graduate and typing	–	clerk
Female and PG degree	–	Secretary
Female, graduate and typing	–	steno.

(b) Create a table result with the following fields:

Reg. No, Name, Mark for three subjects, Total, Result.

(i) Add five records

(ii) Fill total field with the sum of three marks

(iii) Fill the result field with 'pass' if total> = 150 otherwise 'Fail'.

4. (a) Write and test a VB program that allows the user choice among four arithmetic operations of addition, subtraction, multiplication and division with two given numbers.

(b) Create a table pay with the following fields:

Emp. No, Name, Date of joining, Basic pay, DA, HRA, Deductions, Gross pay, Net pay.

(i) Add five records.

(ii) Fill DA with 75% of basic pay.

(iii) Fill HRA with 1500.

(iv) Fill deductions with 2000.

(v) Fill gross pay with the sum of basic pay, DA and HRA.

(vi) Fill net pay with gross pay- deduction.

(vii) Display all the fields.

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5. (a) Write a VB program to do temperature conversion C to F and F to C at user's choice using label, text and enter key.

(b) Create table biodata with the following fields:

Name, Age, Sex, Degree, City.

(i) Add five records

(ii) Display all records of male employees

(iii) Delete all B.Sc. degree holders and then display.

(iv) Display all records having age>20 and city is Trichy.

6. (a) Write and test a VB program to compute and print either the sum of odd numbers or even numbers at the user's choice using label, text and option buttons.

(b) Create table personal with the following fields:

S. No, Name, City, Pin, Phone.

(i) Add five records.

(ii) Display all names without duplicate.

(iii) Display all names in uppercase and initial capital letters.

(iv) Display all names and length of names.

(v) Display name, city in which only beginning portion of name is known.

DE–4360 **37**

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, MAY 2012.

JAVA PROGRAMMING LAB – V

(2007 onwards)

Time : Three hours Maximum : 100 marks

ONE question should be selected by the examiner for each candidate and given to them by lot system.

Each sub-division carries 50 marks

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1. (a) Write a Java Program for sorting a given list of names in ascending order using command line arguments.

   (b) Write a Java Program to illustrates the use of files and Streams.

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2. (a) Write a Java Program to multiply any two order of given matrices.

   (b) Write a Java Program to illustrating the Implement Single inheritance.

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3. (a) Write a Java Program to illustrating the implementation of multiple inheritance using interface.

   (b) To develop the various polygons draw through the Applet code.

4. (a) Write a Java Program to illustrating the implement Hierarchical inheritance.

(b) Write a Java Program to count number of words and characters in a text.

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5. (a) Write a java program to create and manipulate the following Text Areas, Canvas, Scroll bars, Frames, Menus Dialog Boxes.

(b) Write a Java Program to illustrating the implement Single inheritance.

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6. (a) Write create a Java Program to develop the handling mouse events and keyboard events.

(b) Write a Java Program to count number of words and characters in a text.